

Mean Reversion and Momentum Based Proprietary Trading Strategies

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Sussmit Chaudhry¹, Jacob Lebovitz¹, Landon Schofield¹, and Jayant Zhang¹

¹Massachusetts Institute of Technology

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Executive Summary

This report develops and backtests two machine-learning-driven trading strategies — mean reversion and momentum — on 29 DJIA constituents over 2015–2026. Features derived from technical indicators, cross-sectional return ranks, sector momentum, and calendar signals were filtered by Information Coefficient and redundancy pruning, then used to train 120 models across six model types, two strategies, and five holding periods (1 to 126 days). A purged walk-forward validation scheme with 180-day embargo windows was used throughout to prevent data leakage.

Mid-horizon configurations (21–63 days) were the most defensible out-of-sample: the 5-day mean reversion model achieved a Sharpe ratio of 1.34 and CAGR of 19.9% on the test set, while the 63-day momentum model delivered consistent positive risk-adjusted returns across the validation, test, and embargo periods (Sharpe ratios of 1.73, 0.70, and 3.50, respectively). Short-horizon strategies failed due to transaction cost drag, and long-horizon strategies lacked sufficient trades for reliable inference. Realized short-term volatility was the most predictive feature across both strategies, and the two approaches are regime-complementary: mean reversion excels during high-volatility shocks while momentum excels in stable trending markets. A volatility-conditioned allocation combining both strategies is recommended over either approach in isolation.

1 Introduction

This report presents two proprietary trading strategies developed and backtested on the constituents of the Dow Jones Industrial Average over the past 10 years. (WBA was dropped since the company turned private in 2025) The two strategies are based on mean reversion and momentum which represent fundamentally opposing market hypotheses.

Both strategies are implemented as machine learning models trained on cross sectional stock features using walk-forward validation and 180 day trading embargos to preserve temporal relationships and prevent data leakage.

The remainder of this report details the two strategies’ design and rationale, the backtesting framework, and historical performance in addition to a discussion on the key risks and

limitations of this approach.

2 Feature Selection

The feature selection process was designed to reduce noise and prevent overfitting of the model. Different technical features from technical analysis libraries (ta, ta-lib, and pandas-ta) were identified to make a comprehensive list. Indicators were grouped by mean reversion and momentum strategies. For mean reversion, features like RSI¹, Bollinger bands², z-scores, volatility measures, etc. were used. Additionally, for the momentum strategy, features like moving average relationships, MACD, and volatile adjusted momentum were used. All features were calculated using only historical information available at each time. The adjusted prices from Yahoo Finance with OHLCV data was used to make features. The features were standardized using a rolling z-score which is calculated as:

$$Z_{i,t} = \frac{X_{i,t} - \mu_{i,t-1}^{(w)}}{\sigma_{i,t-1}^{(w)}}$$

where $\mu_{i,t-1}^{(w)}$ and $\sigma_{i,t-1}^{(w)}$ are the rolling mean and rolling standard deviation over the previous w observations. The rolling window was shifted by one day so that the value at any time t was scaled only using information available before time t . This helped prevent any look-ahead bias.

The dataset was divided chronologically into training, validation, and test sets with a ratio of 60/15/25% respectively. Different holding periods such as 1-day, 5-day, 21-day, 63-day and 126-day were tested. For each strategy and holding period, different features were filtered out. The Information Coefficient (IC) was used to filter features for each strategy and horizon using the validation data set. Features with an absolute IC value of less than 0.01 were removed. A second filtering method was used to remove extremely correlated features. When two features had a correlation of more than 0.75, the feature with lower absolute IC value was dropped. This was done to eliminate redundancy and reduce noise. The final features were fed into regression and binary classification models for training.

Table 1: Models Used for Training

Regression Models	Classification Models
Ridge Regression	Logistic Regression
Elastic Net Regression	Elastic Net Logistic Regression
Random Forest Regressor	Random Forest Classifier
Histogram Gradient Boosting Regressor	Histogram Gradient Boosting Classifier
XGBoost Regressor	XGBoost Classifier
LightGBM Regressor	LightGBM Classifier

¹Wilder, J. W. (1978). New Concepts in Technical Trading Systems. Trend Research, Greensboro, NC.

²Bollinger, J. (2001). Bollinger on Bollinger Bands. McGraw-Hill, New York. ISBN: 978-0-07-137368-5.

2.1 Data and Feature Construction

All features were computed from Yahoo Finance³ adjusted close prices (OHLCV) for the 29 DJIA-2015 constituents over the period January 2015 to January 2026.⁴ Adjusted close prices account for dividends and stock splits and serve as a proxy for the total return index (TRI). Features were constructed using the `ta`, `ta-lib`, and `pandas-ta` libraries, grouped into six categories:

- **Mean reversion:** z-scores (10/20/50-day), Bollinger Band position and width, RSI (7/14/21-day), Stochastic %K/%D, Williams %R, CCI, VWAP deviation, distance from rolling highs/lows (20d/50d), mean reversion speed (return autocorrelation), price-to-SMA ratios (20/50-day)
- **Momentum:** multi-horizon returns (5d/10d/21d/63d/126d/252d), MACD (line/signal/histogram), SMA crossovers (10/50 and 50/200), ADX and directional indicators, Aroon (25-day), volatility-adjusted momentum (21d/63d), PPO, TSI, CMF, EOM, DEMA/TEMA
- **Volatility:** rolling standard deviation (10d/21d/63d), Average True Range (ATR), short-to-long volatility ratio (10d/63d), Keltner Channel bounds
- **Cross-sectional:** percentile return ranks (21d/63d/126d/252d), RSI rank, volatility rank, excess return versus equal-weight benchmark (21d/63d)
- **Calendar:** sine/cosine encodings of month and quarter (for linear models), month and quarter one-hot dummies (for tree models), month-end/quarter-end flags, January indicator
- **Sector momentum:** equal-weight sector returns (21d/63d/126d), stock-minus-sector relative momentum, sector percentile rank across the universe

The full candidate set comprised 46 features for the mean reversion strategy and 47 for the momentum strategy. All features were constructed using only information strictly available before each prediction date to prevent look-ahead bias.

2.2 Feature Standardization

All features were standardized using a rolling z-score computed on the training window only:

$$Z_{i,t} = \frac{X_{i,t} - \mu_{i,t-1}^{(w)}}{\sigma_{i,t-1}^{(w)}}$$

where $\mu_{i,t-1}^{(w)}$ and $\sigma_{i,t-1}^{(w)}$ are the rolling mean and standard deviation computed over the w observations strictly prior to time t . The one-day lag ensures the scaling parameters are

³Yahoo Finance. Adjusted OHLCV price data. <https://finance.yahoo.com> (accessed via the `yfinance` Python library, April 2026).

⁴WBA was excluded from the universe as the company went private in 2025. DJIA composition remained constant with the 29 companies on the index in 2015 during training. While this is a less realistic universe of stocks because stocks are added and removed from the index over the years, it prevents severe survivorship bias that would occur if we just used the present day makeup of DJIA.

computed exclusively from past data. A **RobustScaler** (based on median and interquartile range rather than mean and variance) was applied as a secondary normalization step to mitigate the influence of outliers, which are common in financial return distributions.

2.3 Train / Validation / Test Split

The dataset was partitioned chronologically into three non-overlapping sets:

- **Training:** April 2016 – February 2022 (39,336 stock-day observations)
- **Validation:** August 2022 – June 2023 (6,322 observations)
- **Test:** December 2023 – December 2025 (9,222–12,847 observations, decreasing for longer horizons)

A 180 trading-day embargo was enforced between each split boundary to prevent information leakage from long-horizon forward returns⁵. Random partitioning was not used, as it would allow future observations to inform model training (look-ahead bias). Within the training set, a purged walk-forward cross-validation scheme (5 folds) was used for model evaluation, with horizon-specific purge gaps applied to prevent target return overlap between folds.

2.4 Feature Selection

Feature selection was conducted exclusively on the validation set to preserve the integrity of the locked test set. Three sequential filters were applied independently for each strategy–horizon combination.

2.4.1 Step 1: Dead Feature Removal

Features with absolute Information Coefficient $|IC| < 0.005$ were unconditionally removed as carrying no statistically distinguishable predictive signal. The IC is the Spearman rank correlation between the feature values and the realized forward return across the cross-section of stocks, averaged over all validation dates⁶:

$$IC = \frac{1}{T} \sum_{t=1}^T \rho_S(X_{\cdot,t}, r_{\cdot,t+h})$$

where ρ_S denotes Spearman rank correlation, $X_{\cdot,t}$ is the vector of feature values across all stocks at time t , and $r_{\cdot,t+h}$ is the corresponding vector of h -day forward returns. Features failing this threshold were most numerous at the 1-day horizon (8–9 eliminated per strategy), consistent with the lower predictability of very short-horizon returns in large-cap equities.

⁵López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley, New York. ISBN: 978-1-119-48208-6.

⁶Grinold, R. C. and Kahn, R. N. (2000). *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk* (2nd ed.). McGraw-Hill, New York. ISBN: 978-0-07-024882-3.

2.4.2 Step 2: Weak Feature Removal

Features with $0.005 \leq |IC| < 0.010$ were classified as weak and removed. This step eliminated an additional 0–7 features per horizon. Longer-horizon targets (63d, 126d) retained nearly all features surviving Step 1, reflecting the higher signal-to-noise ratio at medium-to-long horizons.

2.4.3 Step 3: Redundancy Pruning

Among surviving features, pairs with pairwise Spearman correlation $|r| > 0.75$ were identified. In each pair, the feature with the lower $|IC|$ was dropped. Notable redundancies eliminated include `return_21d` and `roc_21` ($|r| = 1.00$, identical by construction), `sma_10_over_50` and `macd_signal` ($|r| = 0.99$), `vol_adj_mom_63` and `return_63d` ($|r| = 0.987$), and `bb_pband_20` and `zscore_20` ($|r| = 1.00$, both measuring price position relative to a moving average).

2.5 Selection Results

Table 2 summarizes the feature counts after each selection stage.

Table 2: Feature Selection Summary by Strategy and Horizon				
Strategy	Horizon	Candidate	After IC	Final
Mean Reversion	1d	46	31	20
Mean Reversion	5d	46	43	23
Mean Reversion	21d	46	43	22
Mean Reversion	63d	46	45	25
Mean Reversion	126d	46	45	24
Momentum	1d	47	38	24
Momentum	5d	47	44	27
Momentum	21d	47	46	28
Momentum	63d	47	46	31
Momentum	126d	47	46	27

Table 3 reports the top features by absolute IC pooled across horizons and both strategies.

Table 3: Top 15 Features by Absolute IC (Validation Set)

Rank	Feature	Category	IC
1	volatility_10d	Volatility	0.150
2	sector_rank_126d	Sector Momentum	0.140
3	cal_month_sin	Calendar	0.137
4	volatility_21d	Volatility	0.124
5	vol_adj_mom_21	Momentum	0.115
6	sector_rank_63d	Sector Momentum	0.106
7	sma_10_over_50	Momentum	0.104
8	rank_return_126d	Cross-Sectional	0.102
9	cal_month_cos	Calendar	0.099
10	cal_is_month_end	Calendar	0.092
11	aroon_up_25	Momentum	0.091
12	aroon_down_25	Momentum	0.086
13	return_vs_benchmark_63d	Cross-Sectional	0.078
14	return_10d	Momentum	0.062
15	cal_quarter_sin	Calendar	0.061

2.6 Model Training

For each strategy–horizon combination, the selected features were fed into six regression and six binary classification models as listed in Table 4. Regression models predict the raw forward return; binary classification models predict the direction (up/down).

Table 4: Models Used for Training

Regression Models	Classification Models
Ridge Regression	Logistic Regression
Elastic Net Regression	Elastic Net Logistic Regression
Random Forest Regressor	Random Forest Classifier
Histogram Gradient Boosting Regressor	Histogram Gradient Boosting Classifier
XGBoost Regressor	XGBoost Classifier
LightGBM Regressor	LightGBM Classifier

In total, 120 models were trained ($2 \text{ strategies} \times 5 \text{ horizons} \times 12 \text{ models}$). All models were evaluated by IC (regression) and AUC (classification) on the validation set. The best-performing model per strategy–horizon combination was selected to generate trading signals for backtesting.

2.7 Model Performance

Table 5 reports test-set performance averaged across all models, by task and horizon.

Table 5: Average Test-Set Performance by Task and Horizon

Task	Horizon	Mean IC (Test)	Mean AUC (Test)
Regression	1d	0.014	0.505
Regression	5d	0.009	0.498
Regression	21d	0.055	0.522
Regression	63d	0.141	0.564
Regression	126d	0.200	0.593
Binary Classification	1d	0.005	0.500
Binary Classification	5d	-0.015	0.491
Binary Classification	21d	0.019	0.508
Binary Classification	63d	0.172	0.578
Binary Classification	126d	0.183	0.587

Several observations are noteworthy. First, predictive power increases substantially with prediction horizon: regression IC rises from 0.014 at 1d to 0.200 at 126d, and binary AUC rises from 0.500 to 0.587 over the same range. Short-horizon returns (1d, 5d) are effectively unpredictable, consistent with the efficient market hypothesis⁷ for large-cap stocks at high frequency. Second, all regression models showed high overfitting risk in cross-validation (high train-IC, near-zero validation-IC), while classification models showed low overfit risk — suggesting that predicting the *direction* of future returns is a more robust task than predicting the *magnitude*. Third, the best individual configurations on the validation set were:

- **Regression:** Mean reversion strategy, `hist_gbm`, 21-day horizon (val IC = 0.284, test IC = 0.052)
- **Binary classification:** Momentum strategy, `lightgbm_bin`, 63-day horizon (val IC = 0.273, test IC = 0.163, test AUC = 0.567)
- **Best overall test AUC:** Regression at 126d horizon, mean AUC = 0.593

The significant drop in IC from validation to test set in the regression models (e.g., 0.284 → 0.052 for the best mean reversion regression) indicates substantial overfitting despite cross-validation. Classification models exhibited more stable generalization, making them the primary signal source for the backtesting framework.

2.8 Feature Stability Audit

A post-training permutation importance audit assessed feature reliability across horizons using a voting scheme: each feature received 0–3 votes per horizon depending on whether Ridge, Random Forest, and XGBoost models agreed it improved validation-set prediction. Features averaging ≥ 1.5 votes across all five horizons were classified as stable.

The stable core features for each strategy are:

- **Mean Reversion (11 stable features):** `atr_14`, `cal_month_sin`, `cal_is_january`, `sector_mom_63d`, `sector_rank_126d`, `cal_quarter_sin`, `rel_mom_126d`, `bb_width_20`, `cal_month_cos`, `mean_reversion_speed`, `volatility_21d`

⁷Fama, E. F. (1970). "Efficient Capital Markets: A Review of Theory and Empirical Work." The Journal of Finance, 25(2), 383–417

- **Momentum (14 stable features):** rank_return_126d, volatility_21d, cal_month_sin, return_252d, sector_rank_126d, adx_14, sector_mom_63d, cal_month_cos, cal_quarter_sin, rank_return_252d, cal_is_january, rel_mom_126d, sma_50_over_200, adx_pos_14

Five features appear in the stable core of *both* strategies: cal_month_sin, sector_rank_126d, sector_mom_63d, cal_month_cos, and rel_mom_126d. The consistent presence of calendar and sector momentum features across both strategies confirms that these signals capture genuine, strategy-agnostic market structure rather than overfitting to a particular return profile.

3 Backtesting and Performance Analysis

Backtesting was done using the Python library backtrader⁸. Backtrader takes as inputs dates, ticker symbols, and a signal to buy, sell, or hold. It is purely a trading simulator and does not do any feature engineering or model training. Models for both strategies for all holding periods were backtested using Backtrader with the constraint that a strategy could only be executed at most once per holding period that the model was trained on. For example, a mean reversion strategy with a holding period of 5 days could only trade stock positions once every 5 days. Additionally, there was a 10 bps trade execution penalty reflecting the transaction costs that occur from changing positions in the market. Results of the backtesting are shown in Table 6 for the training, validation, testing, and embargo periods.

⁸backtrader. Python backtesting library. <https://www.backtrader.com/docu/> (accessed April 2026).

Table 6: Backtest Results Summary: Bold rows indicate best out-of-sample per strategy on each embargo period. Periods with no gross profit or loss are periods where the position was not closed so there was no realized profit or loss.

Strategy	Horizon	Dataset	Final Value	Return %	CAGR %	Sharpe	Sortino	Calmar	SQN	Max DD %	Max DD Days	Daily Vol %	Ann. Vol %	Trades	Win Rate %	Profit Factor	Payoff Ratio	Gross Profit	Gross Loss
momentum	target_1d	train	2.54M	153.97	17.30	1.21	1.69	1.42	3.72	12.17	257	0.88	13.99	4114	51.60	1.34	1.25	6.13M	4.58M
momentum	target_5d	train	32.51M	3150.53	81.48	4.09	6.62	12.70	6.56	6.42	53	0.94	14.85	1422	67.00	4.20	2.04	40.47M	9.64M
momentum	target_ 21d	train	10.88M	988.41	50.49	2.72	4.07	4.47	7.46	11.29	102	0.97	15.34	268	78.00	5.72	1.40	11.87M	2.08M
momentum	target_ 63d	train	6.43M	543.34	37.53	1.88	2.67	2.35	4.71	15.99	112	1.08	17.11	113	60.20	7.38	4.02	5.70M	771.7K
momentum	target_ 126d	train	2.30M	130.23	15.35	0.82	1.20	0.59	1.39	25.81	287	1.00	15.87	25	56.00	3.07	1.31	1.23M	399.7K
mean_ reversion	target_1d	train	2.29M	129.25	15.26	1.04	1.47	1.36	2.62	11.18	153	0.92	14.62	3197	50.90	1.26	1.21	6.35M	5.04M
mean_ reversion	target_5d	train	47.30M	4629.56	93.52	3.39	5.14	8.34	6.82	11.22	46	1.26	20.04	1218	62.80	4.58	2.67	59.09M	12.91M
mean_ reversion	target_ 21d	train	12.74M	1174.18	54.60	3.18	4.88	7.41	7.03	7.37	37	0.87	13.89	334	72.20	7.89	2.85	13.58M	1.72M
mean_ reversion	target_ 63d	train	7.00M	599.99	39.53	2.33	3.08	2.81	5.21	14.07	137	0.90	14.22	93	72.00	12.35	4.61	4.69M	379.4K
mean_ reversion	target_ 126d	train	2.42M	141.81	16.32	0.84	1.14	0.55	3.10	29.51	115	1.06	16.82	42	66.70	5.02	1.79	1.35M	268.1K
momentum	target_1d	val	853.9K	-14.61	-15.00	-1.14	-1.64	-0.85	-1.68	17.67	244	0.85	13.45	946	39.90	0.78	1.11	525.7K	676.9K
momentum	target_5d	val	891.3K	-10.87	-11.17	-0.71	-1.13	-0.51	-0.80	21.75	134	0.96	15.18	216	43.10	0.79	0.84	450.3K	572.5K

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Table 6 (*continued*)

Strategy	Horizon	Dataset	Final Value	Return %	CAGR %	Sharpe	Sortino	Calmar	SQN	Max DD %	Max DD Days	Daily Vol %	Ann. Vol %	Trades	Win Rate %	Profit Factor	Payoff Ratio	Gross Profit	Gross Loss
momentum	target_ 21d	val	1.29M	29.33	30.28	1.73	2.82	4.48	2.88	6.76	86	1.00	15.83	63	55.60	3.35	2.29	389.0K	116.3K
momentum	target_ 63d	val	1.00M	0.02	0.02	-0.05	0.06	0.00	0.29	7.25	106	0.72	11.39	19	21.10	1.27	1.27	54.4K	43.0K
momentum	target_ 126d	val	1.10M	9.75	10.04	0.87	1.57	1.91	—	5.26	32	0.54	8.59	3	0.00	—	—	0	0
mean_ reversion	target_ 1d	val	987.1K	-1.29	-1.33	-0.02	-0.03	-0.11	-0.26	11.78	226	0.93	14.83	554	44.00	0.95	1.11	442.4K	465.4K
mean_ reversion	target_ 5d	val	1.04M	3.50	3.60	0.30	0.45	0.28	0.40	12.67	168	0.95	15.10	176	43.80	1.12	1.40	410.6K	365.1K
mean_ reversion	target_ 21d	val	1.19M	18.80	19.39	1.58	2.45	2.29	1.19	8.45	83	0.72	11.36	54	51.90	1.70	1.39	324.7K	191.4K
mean_ reversion	target_ 63d	val	1.14M	13.88	14.30	1.14	1.61	1.73	1.72	8.28	64	0.71	11.23	12	41.70	5.57	2.23	45.0K	8.1K
mean_ reversion	target_ 126d	val	1.06M	6.07	6.25	0.42	0.78	1.07	—	5.86	31	0.61	9.72	8	0.00	—	—	0	0
momentum	target_ 1d	test	663.8K	-33.62	-18.97	-1.38	-1.67	-0.45	-2.11	42.05	482	0.91	14.48	1653	44.20	0.73	0.90	909.8K	1.24M
momentum	target_ 5d	test	1.05M	5.36	2.74	0.25	0.38	0.20	0.28	13.52	229	0.94	14.91	492	47.20	1.06	1.08	952.5K	897.5K
momentum	target_ 21d	test	834.3K	-16.57	-9.24	-0.66	-0.79	-0.37	-0.90	24.73	406	0.87	13.85	82	47.60	0.71	0.64	428.9K	602.6K
momentum	target_ 63d	test	1.21M	21.39	12.06	0.70	1.04	0.71	1.13	17.05	147	1.03	16.42	34	41.20	1.99	1.28	235.3K	118.3K
momentum	target_ 126d	test	1.02M	2.13	1.46	0.00	0.19	0.09	-0.62	15.93	129	0.92	14.58	6	16.70	0.39	0.39	5.7K	14.5K
mean_ reversion	target_ 1d	test	753.8K	-24.62	-13.50	-0.82	-1.06	-0.47	-1.61	29.00	464	1.02	16.17	1102	49.70	0.82	0.82	1.10M	1.34M

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Table 6 (continued)

Strategy	Horizon	Dataset	Final Value	Return %	CAGR %	Sharpe	Sortino	Calmar	SQN	Max DD %	Max DD Days	Daily Vol %	Ann. Vol %	Trades	Win Rate %	Profit Factor	Payoff Ratio	Gross Profit	Gross Loss
mean_ reversion	target_5d	test	1.42M	41.98	19.89	1.34	1.97	2.03	2.28	9.79	118	0.90	14.24	370	56.20	1.47	1.10	1.38M	939.9K
mean_ reversion	target_21d	test	842.6K	-15.74	-8.76	-0.44	-0.46	-0.32	-1.11	27.40	445	1.13	17.93	72	43.10	0.66	0.81	353.3K	537.9K
mean_ reversion	target_63d	test	980.2K	-1.98	-1.17	-0.06	0.02	-0.07	-0.37	17.40	203	1.07	16.99	30	26.70	0.78	1.27	171.2K	218.6K
mean_ reversion	target_126d	test	1.37M	37.13	24.28	0.99	2.50	2.40	—	10.10	75	1.36	21.57	3	33.30	—	—	347.5K	0
momentum	target_1d	embargo_ train_val	794.9K	-20.51	-37.51	-2.62	-3.38	-1.36	-2.62	27.60	117	1.10	17.42	311	39.20	0.56	0.86	272.7K	483.3K
momentum	target_5d	embargo_ train_val	983.5K	-1.65	-3.36	-0.16	-0.20	-0.25	-0.16	13.38	118	0.96	15.17	124	46.00	0.95	0.96	310.0K	327.1K
momentum	target_21d	embargo_ train_val	897.4K	-10.26	-19.89	-2.54	-3.11	-1.47	-1.15	13.49	87	0.55	8.79	26	42.30	0.51	0.51	68.9K	134.7K
momentum	target_63d	embargo_ train_val	900.2K	-9.98	-19.38	-0.87	-0.92	-0.88	—	21.95	39	1.46	23.19	1	0.00	—	—	0	0
momentum	target_126d	embargo_ train_val	1.00M	0.00	0.00	—	—	—	—	0.00	0	0.00	0.00	0	—	—	—	0	0
mean_ reversion	target_1d	embargo_ train_val	952.0K	-4.80	-9.58	-0.41	-0.62	-0.60	-0.87	16.08	117	1.25	19.83	275	52.00	0.84	0.76	383.2K	457.4K
mean_ reversion	target_5d	embargo_ train_val	858.8K	-14.12	-26.79	-1.59	-2.12	-1.19	-1.73	22.47	118	1.18	18.68	113	43.40	0.58	0.67	203.4K	347.5K
mean_ reversion	target_21d	embargo_ train_val	944.3K	-5.57	-11.08	-0.75	-0.87	-0.78	-1.51	14.23	69	0.94	14.89	30	36.70	0.45	0.62	45.0K	99.4K
mean_ reversion	target_63d	embargo_ train_val	979.5K	-2.05	-4.15	-0.07	-0.02	-0.19	—	21.40	60	1.70	26.92	1	0.00	—	—	0	0

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Table 6 (*continued*)

Strategy	Horizon	Dataset	Final Value	Return %	CAGR %	Sharpe	Sortino	Calmar	SQN	Max DD %	Max DD Days	Daily Vol %	Ann. Vol %	Trades	Win Rate %	Profit Factor	Payoff Ratio	Gross Profit	Gross Loss
mean_ reversion	target_ 126d	embargo_ train_val	1.00M	0.00	0.00	—	—	—	—	0.00	0	0.00	0.00	0	—	—	—	0	0
momentum	target_1d	embargo_val_ test	1.01M	0.81	1.68	0.20	0.26	0.28	-0.72	6.10	69	0.71	11.35	346	43.40	0.85	0.97	191.1K	225.3K
momentum	target_5d	embargo_val_ test	1.10M	10.18	22.18	2.08	4.08	3.96	1.04	5.60	66	0.62	9.84	126	45.20	1.43	1.08	179.7K	125.5K
momentum	target_ 21d	embargo_val_ test	932.1K	-6.79	-13.53	-1.48	-1.78	-1.26	-1.30	10.69	92	0.62	9.83	15	20.00	0.37	1.25	35.2K	94.0K
momentum	target_ 63d	embargo_ val_test	1.12M	11.76	25.81	3.50	3.90	8.30	—	3.11	21	0.40	6.28	3	0.00	—	—	0	0
momentum	target_ 126d	embargo_val_ test	1.00M	0.00	0.00	—	—	—	—	0.00	0	0.00	0.00	0	—	—	—	0	0
mean_ reversion	target_1d	embargo_val_ test	870.4K	-12.96	-24.94	-2.39	-3.20	-1.83	-1.95	13.64	116	0.74	11.78	300	43.70	0.58	0.74	126.3K	216.7K
mean_ reversion	target_5d	embargo_val_ test	956.7K	-4.33	-8.74	-0.83	-1.26	-0.70	-0.39	12.41	86	0.66	10.53	92	45.70	0.85	1.00	181.4K	212.4K
mean_ reversion	target_ 21d	embargo_val_ test	962.0K	-3.80	-7.68	-0.83	-0.96	-0.83	-0.48	9.26	72	0.61	9.66	24	50.00	0.71	0.41	94.6K	133.7K
mean_ reversion	target_ 63d	embargo_ val_test	1.10M	9.95	21.65	2.96	4.67	13.11	—	1.65	5	0.40	6.29	7	0.00	—	—	0	0
mean_ reversion	target_ 126d	embargo_val_ test	1.00M	0.00	0.00	—	—	—	—	0.00	0	0.00	0.00	0	—	—	—	0	0
momentum	target_1d	all	1.11M	11.02	1.08	0.14	0.20	0.02	0.20	62.40	975	0.89	14.17	7341	47.90	1.01	1.09	9.80M	9.69M
momentum	target_5d	all	33.04M	3204.09	43.32	2.52	3.89	1.73	3.56	25.08	369	0.93	14.69	2333	61.90	1.49	0.92	99.11M	66.63M
momentum	target_ 21d	all	17.06M	1605.70	34.15	2.07	3.15	2.58	5.36	13.26	341	0.91	14.52	482	68.70	2.24	0.97	28.33M	12.67M

Continued on next page

Table 6 (*continued*)

Strategy	Horizon	Dataset	Final Value	Return %	CAGR %	Sharpe	Sortino	Calmar	SQN	Max DD %	Max DD Days	Daily Vol %	Ann. Vol %	Trades	Win Rate %	Profit Factor	Payoff Ratio	Gross Profit	Gross Loss
momentum	target_ 63d	all	7.01M	601.09	22.78	1.23	1.78	0.72	2.38	31.83	492	1.06	16.86	176	60.20	1.95	1.16	9.14M	4.70M
momentum	target_ 126d	all	2.26M	125.66	9.21	0.50	0.82	0.36	1.07	25.81	649	0.93	14.81	54	68.50	1.79	0.78	2.16M	1.21M
mean_ reversion	target_ 1d	all	1.40M	40.43	3.55	0.30	0.42	0.08	0.62	44.19	1000	0.96	15.17	5405	50.00	1.04	1.04	10.61M	10.21M
mean_ reversion	target_ 5d	all	53.05M	5205.30	50.48	2.33	3.54	2.19	4.10	23.08	500	1.15	18.24	1906	58.40	1.55	1.10	147.92M	95.54M
mean_ reversion	target_ 21d	all	16.74M	1573.88	33.89	2.14	3.06	1.97	3.85	17.17	391	0.88	13.93	549	64.50	1.99	1.06	31.46M	15.83M
mean_ reversion	target_ 63d	all	7.08M	607.62	22.90	1.38	1.91	0.69	1.93	33.46	895	0.94	14.90	158	60.80	2.03	1.10	10.03M	4.95M
mean_ reversion	target_ 126d	all	4.44M	344.03	17.51	0.92	1.32	0.59	4.35	29.51	502	1.03	16.41	66	69.70	6.02	1.96	3.16M	524.8K

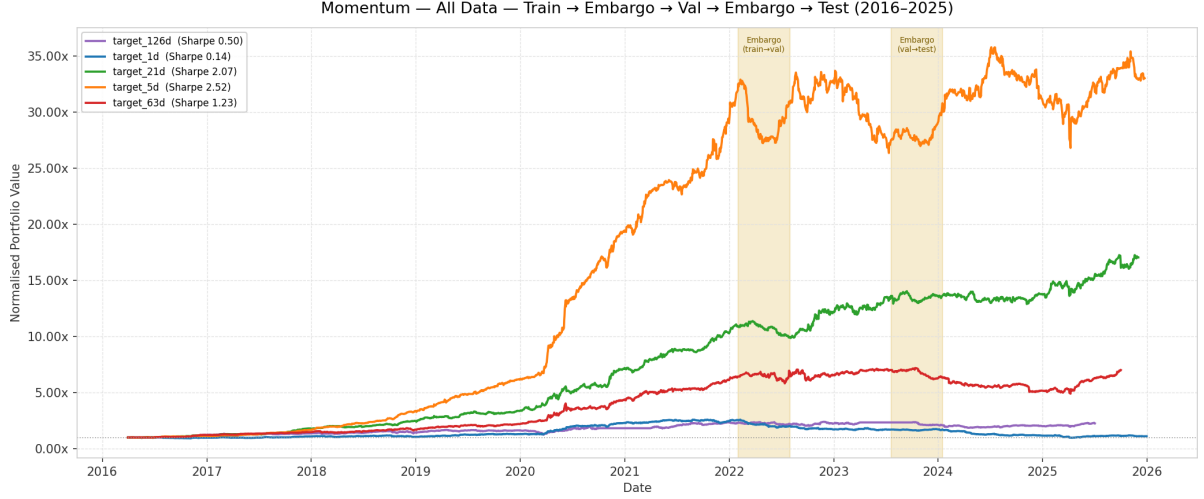


Figure 1: Momentum strategy equity curves across all dataset segments (2016–2025) for each holding period. Shaded bands mark the 180-day embargo windows separating train/validation and validation/test. The flattening of the orange (5-day) and green (21-day) curves after the training period illustrates the degradation of in-sample performance out-of-sample.

3.1 In-Sample vs. Out-of-Sample Performance

There is a significant difference between the in-sample and out-of-sample performance of the two strategies. The 5-day mean reversion model had a CAGR of 93.5% and Sharpe ratio of 3.39 in-sample but a Sharpe ratio of 1.34 on the test set and negative Sharpe ratio on the embargo periods. The 5-day momentum model showed even worse performance decay out-of-sample with a Sharpe ratio of 4.09 in-sample and a -0.71 and 0.25 Sharpe ratio on the validation and test sets respectively. This performance degradation can be attributed to the fact that it is very difficult to predict future prices with past signals that were likely regime specific. See Figures 1 and 2 for performance across the 10 year period.

We interpret this gap between in-sample and out-of-sample performance as an indication of overfitting. There were feature-selection filters in place during training with information coefficient screening removing features with low predictive power. However, the complexity of some of the tree based models looks to have fit patterns in the data that fail to generalize well. Nevertheless, the mid-horizon strategies do have positive risk-adjusted returns on at least one out-of-sample set of data which suggests that the signals predicted by the model are not completely useless. Furthermore, we are using a the DJIA list from 2015 to eliminate the survivorship bias that would exist if we used companies from 2026. The DJIA is a specific list of companies that is meant to capture the economy of the United States. In our analysis, however, the models' poor performance on future data could also be attributed to the fact that the 30 companies from 2015 do not necessarily reflect the DJIA or U.S. economy of 15+ years into the future.

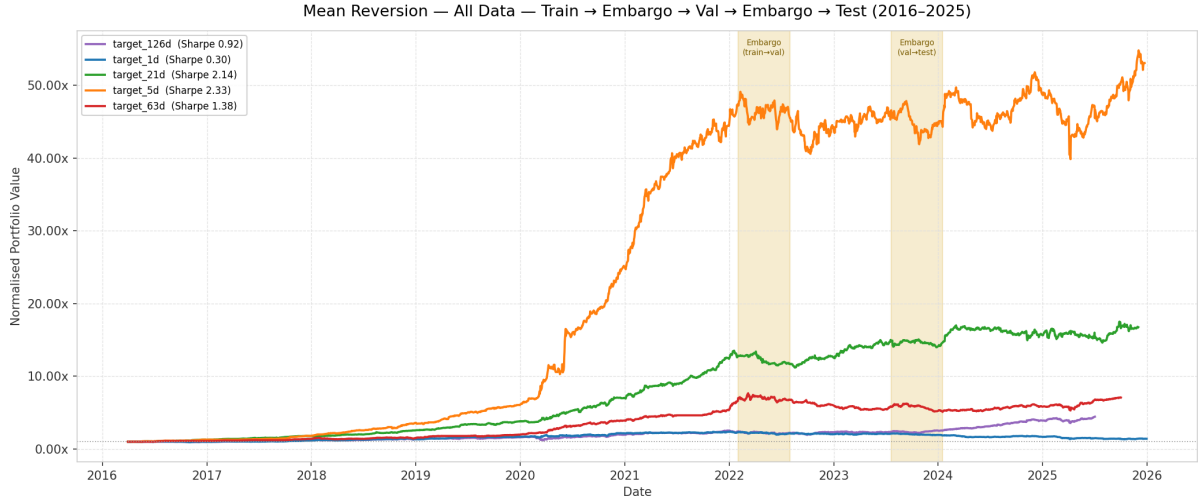


Figure 2: Mean reversion strategy equity curves across all dataset segments (2016–2025) for each holding period. Shaded bands mark the 180-day embargo windows. The 5-day horizon achieves the largest in-sample gain but stalls almost entirely after 2022, consistent with the hypothesis that short-horizon reversal signals are regime-dependent.

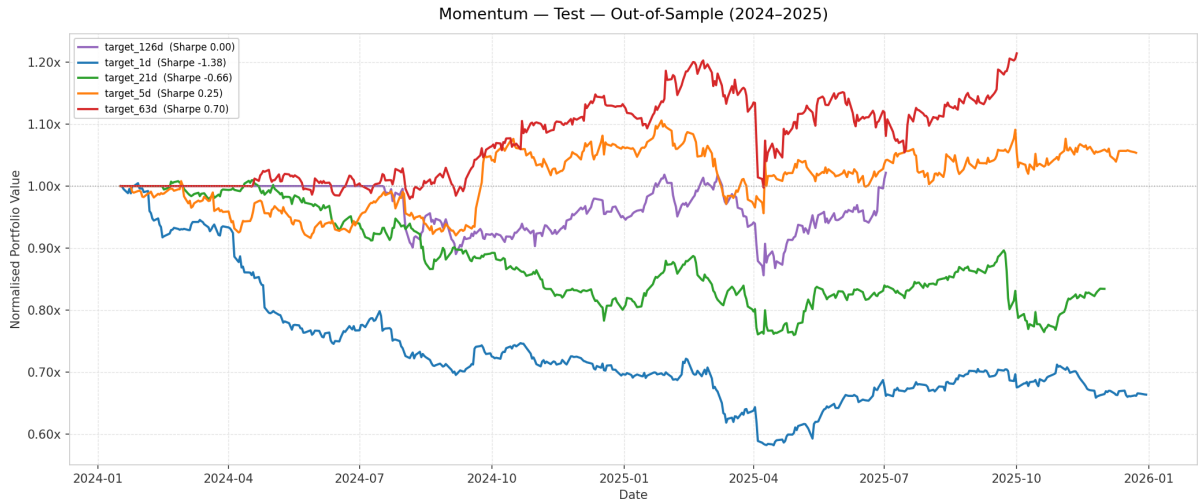


Figure 3: Out-of-sample equity curves for the momentum strategy on the held-out test period (2024–2025). The 63-day model (red) outperforms all other horizons, while the 1-day model (blue) loses over a third of its starting value. The divergence across horizons motivates the conclusion that mid-horizon momentum is the most defensible configuration for deployment.

3.2 Horizon Sensitivity

The holding period/trading frequency has a significant impact on the performance of the strategies. Both mean reversion and momentum strategies with 1-day horizons underperformed on out-of-sample data. Momentum returns -33.6% on the test set and mean reversion returns -24.6% with maximum drawdowns of 42% and 29% respectively. There are two main reasons for the poor performance. Firstly, the signal to noise ratio is very low in comparison to the 10 bps per trade transaction cost in the simulator. Secondly, the number of trades was very high in the out of the sample periods which causes transaction costs to compound over time.

At the high end of the horizons, 126 days, the strategies suffer from lack of data. The 180-day embargo is the length of the holding period which produces 0 executed trades during those periods. We cannot have reliable statistical inference on these samples, so the 37.1% returns after 3 trades is more anecdotal than anything.

The mid-range horizons of 21-days and 63-days seem to perform the best. Mean reversion with a 21-day holding period produces a Sharpe ratio of 1.58, and the 63-day model performed the best on the embargo period between validation and test sets with a Sharpe ratio of 2.96. As Figure 3 illustrates, momentum at 63 days performed the best in the test set with a Sharpe ratio of 0.7 and an embargo period Sharpe ratio of 3.50. These two horizons appear long enough to filter out noise but short enough to trade frequently to get statistically meaningful results.

3.3 Strategy Comparison

As the theory dictates, the mean reversion strategy performs better than the momentum strategy on shorter horizons while the momentum strategy performs better on longer horizons. The best out-of-sample mean reversion performance on the test set was the 5-day model with a CAGR of 19.9% and Sharpe ratio of 1.34. The best momentum performer was the 63-day model with a CAGR of 12.1% and Sharpe ratio of 0.7. This trend matches with intuition: mean reversion exploits short-term price deviations from the mean where momentum captures persistent long-term trends.

Neither strategy was consistent in being profitable across all out-of-sample periods. This actually presents a useful insight which is that a portfolio of both strategies would perform better than either strategy in isolation.

3.4 Embargo Period Performance

The embargo periods are treated as independent out-of-sample tests. Performance in these windows is generally worse than in the validation and test set windows which is somewhat expected. Even small shifts in time horizon should degrade model performance. The notable exception is the 63-day momentum model which has a 25.8% CAGR and Sharpe ratio of 3.50. Even though there were only 3 trades during this period, the consistent good performance of this model on the validation, test, and embargo period suggest that we did capture a modest signal that is less sensitive to temporal shifts.

3.5 Risk Characteristics

Across all strategies, the daily return volatility is stable from 0.6-1.4% and annualized volatility ranging from 10-22%. Maximum drawdowns are where we see differences. The 1-day momentum model drew down 42% during the test set whereas we only saw 9.8% drawdown for the 5-day mean reversion model. Shorter horizons are penalized by frequent trading because of the transaction costs.

The system quality number (SQN) values further highlight this point. SQN can be thought of as a t-test of sorts. Only mid-horizon strategies have positive SQN values with none of the out-of-sample testing generating $SQN > 2.5$ which is typically used as a threshold for a “good” model. It should be noted that some strategies do not have an SQN number because they did not close out their positions in the simulations for the period. In other words, they were still in the market by the end of the period.

4 Strategy Analysis and Macroeconomic Context

4.1 Implications for Deployment

There are three main takeaways from the backtesting. The first is that the transaction costs are a binding constraint. 10 bps transaction costs cause high frequency trading strategies to underperform out-of-sample. While 10 bps may be a high transaction cost, it highlights the fact that transaction costs are indeed real and can be a major reason why high frequency trading strategies fail. The 21-day and 63-day horizon models’ performance show that they could be candidates for deployment after further validation and testing. With the results we have now, we would not want to commit capital without more fine-tuning.

4.2 Strategy Rationale

The two strategies exploit fundamentally opposing market hypotheses. Mean reversion assumes that asset prices temporarily deviate from their fair value due to overreaction, forced selling, or liquidity shocks, and that these deviations are subsequently corrected^{9,10}. Momentum assumes that assets exhibiting strong recent performance will continue to outperform due to persistent investor behavior, gradual information diffusion, and institutional herding¹¹. The two hypotheses are not mutually exclusive — they operate on different timescales and under different market regimes — which is why both were implemented and tested across multiple horizons.

4.3 The Role of Volatility

Volatility is the single most important feature in the pipeline, with `volatility_10d` achieving the highest IC of 0.150 across all features and both strategies. However, its

⁹Poterba, J. M. and Summers, L. H. (1988). "Mean Reversion in Stock Prices: Evidence and Implications." *Journal of Financial Economics*, 22(1), 27–59.

¹⁰De Bondt, W. F. M. and Thaler, R. (1985). "Does the Stock Market Overreact?" *The Journal of Finance*, 40(3), 793–805.

¹¹Jegadeesh, N. and Titman, S. (1993). "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency." *The Journal of Finance*, 48(1), 65–91.

relationship with the two strategies is directionally opposite, and understanding why is central to interpreting the backtesting results.

4.3.1 Why High Volatility Favors Mean Reversion

Mean reversion strategies profit when prices overshoot their equilibrium and subsequently correct. High realized volatility is precisely the environment in which this overshooting is most severe and most reliably reversed. During episodes of elevated volatility, institutional investors face margin calls, risk limits are breached, and forced selling creates temporary dislocations that are unrelated to fundamental value. These price dislocations are the raw material of mean reversion: the larger the deviation, the stronger the expected correction, and the cleaner the signal.

Formally, the mean reversion z-score feature captures this dynamic:

$$Z_{i,t} = \frac{P_{i,t} - \mu_{i,t}^{(w)}}{\sigma_{i,t}^{(w)}}$$

where a large absolute z-score indicates that the price has moved far from its recent average — precisely the condition under which a reversal is most likely. High volatility inflates $\sigma_{i,t}^{(w)}$, but it inflates the numerator even more during shock episodes, producing extreme z-scores that have historically reverted. The empirical confirmation from the backtest is stark: the 5-day mean reversion model returned +36.5% in March 2020, +21.6% in April 2020, and +50.1% in June 2020 — the three highest-volatility months of the decade — accounting for a disproportionate share of the strategy’s entire 10-year compounded return.

4.3.2 Why High Volatility Hurts Momentum

Momentum strategies profit when trends persist long enough to be captured within the holding period. High volatility is structurally hostile to this because it introduces sharp, unpredictable reversals that cut trends short. A momentum signal generated on day t — for example, a stock that has outperformed over the prior 63 days — is rendered useless if a macro shock on day $t + 1$ abruptly reverses sector leadership. The higher the volatility, the shorter the half-life of any trend, and the greater the probability that the momentum signal will be stale by the time it is acted upon.

This is captured in the vol-adjusted momentum feature:

$$\text{vol_adj_mom}_{i,t}^{(h)} = \frac{r_{i,t}^{(h)}}{\sigma_{i,t}^{(w)}}$$

where scaling by volatility deliberately downweights momentum signals in high-volatility environments, acknowledging that raw momentum is less reliable when markets are turbulent. The IC of `vol_adj_mom_21` ($|IC| = 0.115$) significantly exceeds that of the raw `return_21d` feature, validating this adjustment empirically.

4.3.3 The Volatility Regime Threshold

The practical implication is that volatility acts as a natural regime indicator for strategy selection. When realized 10-day volatility is elevated — corresponding to market stress, geopolitical shocks, or forced-selling episodes — mean reversion signals are stronger and more reliable. When volatility is low and stable, trends persist longer, momentum signals have higher predictive power, and mean reversion signals are dominated by noise. Table 7 summarizes this relationship.

Table 7: Volatility Regime and Strategy Signal Quality

Regime	Mean Reversion Signal	Momentum Signal
Low volatility (stable trend)	Weak — noise dominates	Strong — trends persist
High volatility (shock/crisis)	Strong — overshoots revert	Weak — trends cut short
Transition (regime change)	Unreliable	Unreliable

4.4 Macroeconomic Regime Analysis

The backtesting results across 2016–2025 provide a natural experiment in how different macroeconomic regimes affect the two strategies. Four distinct periods are identifiable in the data.

4.4.1 2017–2019: Low Volatility Bull Market

The 2017–2019 period was characterized by the post-tax-reform expansion, moderate Fed tightening, and sustained cross-sectional dispersion in equity returns. Realized volatility was consistently below its long-run average, sector leadership was stable, and price trends were persistent across quarters. This environment was structurally favorable for momentum: the 5-day momentum model produced a sequence of positive months from 2017 through mid-2019, while mean reversion signals were weak because prices rarely strayed far enough from their means to generate high-confidence reversal signals. The monthly heatmap in Figure 4 confirms this: momentum delivers consistent single-digit positive returns across most months of 2017–2019, while mean reversion is more erratic in that period.

4.4.2 2020: COVID Shock and the Mean Reversion Windfall

The COVID-19 pandemic produced the largest volatility spike in the sample, with the VIX reaching 82 in March 2020. The combination of forced selling, liquidity hoarding, and correlated margin calls created price dislocations across all 29 DJIA constituents simultaneously. This was the ideal environment for mean reversion: prices fell far below any reasonable estimate of fair value, z-scores reached multi-sigma extremes, and the subsequent recovery was both rapid and large. The 5-day mean reversion model captured this with returns of +36.5%, +21.6%, and +50.1% in March, April, and June 2020 respectively. The momentum strategy, by contrast, suffered during the same period: the rapid reversal from the drawdown made prior momentum signals obsolete within days.

4.4.3 2022: Fed Tightening and Regime Transition

The Federal Reserve’s pivot from near-zero rates to the fastest tightening cycle since the 1980s — raising the federal funds rate from 0.25% to 4.5% within twelve months — created a regime transition that damaged both strategies. For momentum, the abrupt reversal in sector leadership from growth to value made trailing signals unreliable: stocks that had outperformed in 2021 became the worst performers in 2022. For mean reversion, the persistent downtrend in equities meant that extreme z-scores were not reliably reverted; in a true bear market, oversold conditions can remain oversold for extended periods. The backtesting results reflect this: both strategies underperformed in 2022, with the mean reversion model generating negative Sharpe ratios on the embargo period between training and validation.

4.4.4 2023: Choppy Market and Factor Rotation

The 2023 environment was characterized by rapid rotation between sectors driven by alternating inflation and soft-landing narratives, combined with an AI-driven rally concentrated in a small number of mega-cap technology stocks — most of which (NVIDIA, Meta) are not constituents of the DJIA-2015 universe. This produced a market where both strategies struggled: trends were too short-lived for momentum to capture profitably, and mean reversion signals generated by the broad market failed to account for the concentrated nature of returns. The monthly heatmap shows near-zero and negative months for both strategies across much of 2023.

4.4.5 2024–2025: Persistent Trends and the Tariff Shock

The 2024–2025 period was dominated by persistent megacap-led trends punctuated by sharp but short-lived shocks. The 63-day momentum model performed best in this environment, achieving a Sharpe ratio of 0.70 on the test set and 3.50 on the embargo period between validation and test — the strongest out-of-sample result across all strategy-horizon combinations. The 63-day horizon was long enough to capture the persistent sector trends, while short enough to avoid being overwhelmed by the sharp reversals. The April 2025 tariff-driven shock provided a brief window for mean reversion — the 5-day model recovered within two months — but the broader environment remained trend-friendly, consistent with the momentum model’s relative outperformance.

4.5 Strategy Selection Framework

The macroeconomic analysis and backtesting results together motivate a regime-conditional approach to strategy deployment. Rather than committing to a single strategy, the evidence suggests that allocation between mean reversion and momentum should be conditioned on observable market state variables. Table 8 formalizes this framework.

Table 8: Regime-Conditional Strategy Selection Framework

Market Regime	Preferred Strategy	Preferred Horizon	Observable Indicators
Crisis / forced selling	Mean Reversion	5d – 21d	VIX > 30, extreme z-scores, high cross-sectional dispersion
Stable bull market	Momentum	21d – 63d	VIX < 20, persistent sector leadership, low factor rotation
Regime transition	Reduce exposure	All	VIX 20–30, rapid factor rotation, conflicting signals

The practical implementation of this framework would involve monitoring realized volatility as the primary regime indicator, with VIX or realized 10-day volatility serving as the switch between strategy allocations. A portfolio that tilts toward mean reversion when `volatility_10d` is in the top quartile of its historical distribution, and toward momentum when it is in the bottom quartile, should outperform either standalone strategy on a risk-adjusted basis — a hypothesis supported by the complementary performance profiles observed across the 2016–2025 backtesting period.

The key limitation of this framework is that regime transitions are only clearly identifiable in hindsight. The 2022 tightening cycle and the 2023 factor rotation both appeared gradual at first and sharp in retrospect, making real-time regime detection a non-trivial problem. Further work on dynamic regime classification — for example, using Hidden Markov Models on volatility and dispersion signals — would be a natural extension of the current approach.

The monthly return heatmap in Figure 4 makes clear that neither strategy is regime-agnostic, and the question of which to deploy depends heavily on the prevailing market environment. Momentum is a trend-exploitation strategy and therefore performs best when cross-sectional dispersion is high and winners persist: the 2017–2019 steady bull market and the extended 2020–2021 recovery both produced long sequences of positive months for the 5-day momentum model, with only isolated drawdowns. In contrast, momentum is structurally vulnerable to sharp regime changes — the strategy lost ground in March–April 2022 (9.9% and 5.9%) when the Fed’s pivot to tightening abruptly reversed the prior year’s leadership, and it again struggled through the choppy 2023 tape, where factor leadership rotated rapidly and trends did not to last long enough to be captured.

Mean reversion, by contrast, thrives precisely in the conditions that hurt momentum: high realized volatility, wide intraday and weekly dispersion, and forced-selling episodes where prices overshoot fundamentals. This is most visible in the 2020 COVID shock, where the 5-day mean reversion model returned +36.5% in March, +21.6% in April, and +50.1% in June — a single quarter that accounts for a meaningful share of the strategy’s entire 10-year compounded return — and again during the April 2025 tariff-driven drawdown, which the model recovered from within two months. The flip side is that mean reversion fails in low-volatility, low-dispersion environments such as 2023, where reversal signals are dominated by noise and the 10bps transaction cost erodes small edges; the heatmap shows a string of near-zero and negative months for mean reversion across that year. In the 2024–2025 out-of-sample period, the 63-day momentum model outperformed the

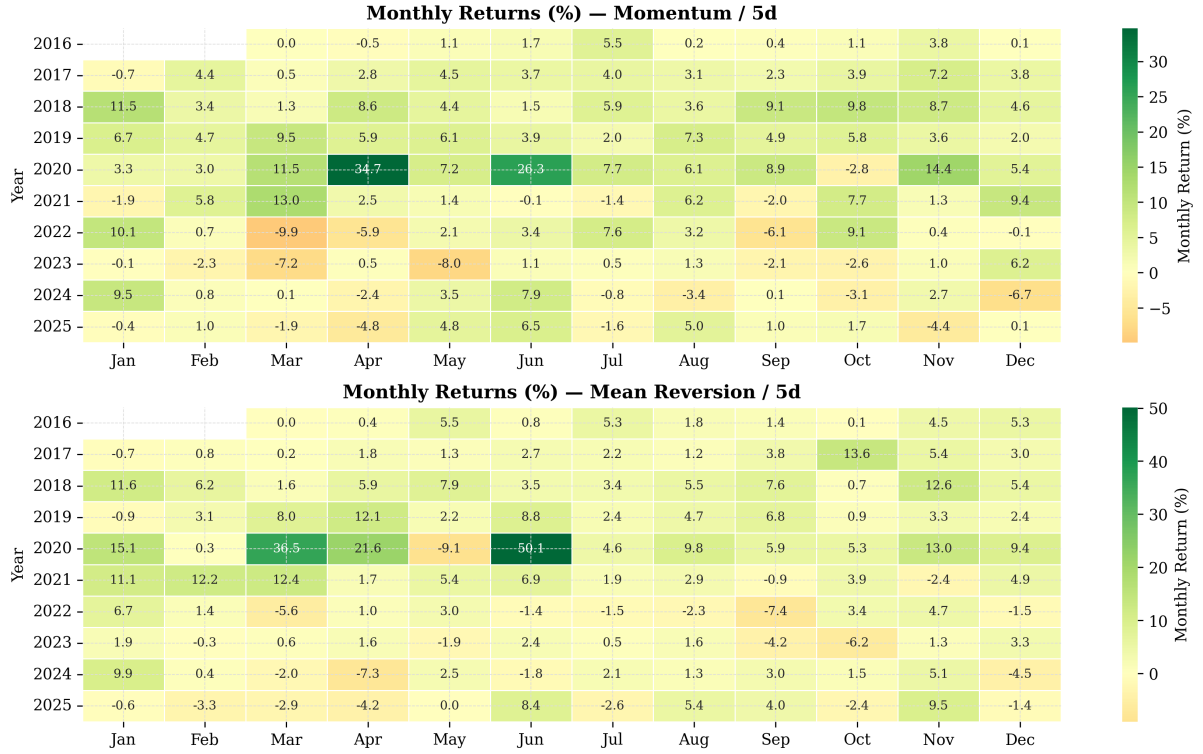


Figure 4: Monthly returns (%) for the best-performing 5-day momentum (top) and 5-day mean reversion (bottom) models across the full 2016–2025 period. The heatmap highlights the regime-dependence discussed previously: mean reversion dominates during high-volatility shock episodes (Mar–Jun 2020, mid-2025), while momentum delivers more consistent returns during stable trending years (2017–2019) and suffers in regime-transition years (2022, 2023).

shorter-horizon mean reversion model, consistent with a market that was driven by persistent megacap-led trends punctuated by sharp but short-lived shocks, which favors longer-horizon trend capture.

The practical implication is that the two strategies are complementary rather than substitutes: a combined book or dispersion-conditioned allocation that tilts toward mean reversion when realized volatility is elevated and toward momentum when trends are stable should dominate either standalone strategy reduce the regime risk that has penalized each individually.

5 Conclusion

Across the ten-year 2016–2025 sample of DJIA constituents, we trained and backtested two families of machine-learning trading strategies—one grounded in mean reversion, the other in cross-sectional momentum—over five holding horizons (1d, 5d, 21d, 63d, 126d) with purged walk-forward validation, 180-day embargoes, and a 10 bps round-trip transaction cost. Here are our major findings throughout this analysis and comparison of different strategies:

1. The mid-horizon band (21–63 days) is the only regime where either strategy survives the transaction-cost hurdle out-of-sample. At the 1-day horizon, both strategies generated negative Sharpe ratios on the test set (−1.38 momentum, −0.82 mean reversion) despite in-sample Sharpes above 1.0, as the signal-to-noise ratio collapsed below the 10 bps trading friction. At the 126-day horizon, the 180-day embargo and 2-year test window left too few executed trades (0–6) for any statistically meaningful inference. The 21-day and 63-day models were the only configurations to deliver positive risk-adjusted returns on at least one out-of-sample window.

2. Mean reversion and momentum are complementary, not substitutable, rather they fail in orthogonal regimes. The monthly-return heatmap (Figure 4) shows mean reversion delivering its largest contributions during high-volatility dislocations (+36.5% / +21.6% / +50.1% in Mar–Jun 2020; recovery inside two months after the April 2025 tariff shock) and failing in the low-dispersion 2023 tape, while momentum produced its most consistent record through the trending 2017–2019 and 2020–2021 markets and was worsened by the 2022 Fed-pivot reversal and the 2023 leadership rotation. This structural negative correlation of failure modes is the most actionable finding of the project.

6 Attribution

The Claude Opus 4.6 model is one of the strongest models for planning and thinking. Therefore, it was used to brainstorm ideas, discuss feature selection methods like IC feature filtering process. Other strategies like PCA, filtering after training, etc were also discussed. Using correct time based splitting of dataset, and scaling the dataset is extremely important for this project so extensive discussion was done on these aspects of the project. A detailed plan was developed for each part of the project. Codex 5.4 model has a huge context window of 1 million tokens and therefore can store huge memory and context for coding tasks while preventing hallucinations. Therefore, it was used to

implement the plan and for coding ML models with relevant plots. The code was manually verified with print statements and plots.

Claude Opus 4.6 was used for writing the backtesting code and producing the plots of backtesting results. Since the models were already trained and the results of the validation and testing were saved in .csv files, the LLM was used to write the code to load the signals into backtrader and have backtrader execute.

Claude Sonnet 4.6 was also used to generate an outline for how to present the findings from the backtesting. It was not used to generate the exact words in the report. However, it was used to generate the executive summary to ensure all relevant information was included.

Specific Prompts Used:

- "What are the best python packages to build technical indicators?"
- "What metrics can be used for filtering features for prop-trading? Is filtering before or after training better?"
- "What are good AUC scores and Spearman correlations in finance?"
- "What are the best ways to split and scale data for prop-trading project? I need to make sure there is no data leakage."
- "Give me ideas for visualizations that would clearly demonstrate the differences in success/behavior for each strategy (mean reversion and momentum, and explain how each would be interpreted."
- "Given the data on the backtesting results, what economic and macro-level events were happening at the time that could have indicated or caused mean reversion or momentum strategies to have more success"
- "Can you [make] something that will show the results [of the backtesting] over time? or a way to show more of the time in relation, rather than just comparing the train/val/test sets?"
- "I am tasked with acting as a trader and analyzing the strategies, specifically looking at backtesting results. Look at the backtesting py file which should execute the backtest for the strategies, as well as some initial results in the csv. Give me another file that will generate insightful and powerful analysis that will compare the different strategies"
- "Write me a python script that loads the models and uses backtrader to backtest the models' performance. Include common summary statistics. Test and validation test are included in backtest_handoff folder. Use the last cell of the .ipynb to get details on how backtrader works. Ensure that the trader is only allowed to trade in accordance with that model's horizon"
- "Create meaningful plots that show the results of backtesting the strategies. Specifically highlight the embargo periods to draw attention to those periods."